Beyond 3D Convolutions: A 4D Spatiotemporal Transformer-Based Network for Capturing Vascular Kinetics in Neoadjuvant Chemotherapy Response Prediction

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Abstract. Neoadjuvant chemotherapy (NAC) is commonly administered before surgery in patients with advanced breast cancer, particularly those with nodal involvement, large tumors, or aggressive biologic subtypes such as inflammatory, HER2-positive, and triple-negative disease. Despite the well-established benefits of NAC, many patients do not respond to treatment, highlighting the need for improved response prediction. Delays introduced by ineffective NAC may contribute to decreased treatment success rates in addition to the common adverse effects of chemotherapy. Although deep learning models have been proposed to predict patient response to NAC, accuracy is often limited by unavailable clinical information to contextualize MRI features, reliance on expert-engineered features beyond tumor masks, and the disregard or simplification of temporal dynamics of the administered contrast. We propose a new 4D transformer-based architecture to predict NAC response from variable length pre-treatment dynamic contrast enhanced MRI (DCE-MRI) sequences. Unlike other state-of-the-art models, we use a deep patch embedding network that leverages spatiotemporal multi-head self-attention to map the physiological progression of contrast agent washout. By capturing complex perfusion dynamics, the proposed 4D attention mechanism transforms dynamic anatomical information into enriched representations that improve the predictions of complete response to NAC. Our model was trained and evaluated on a public dataset with pre-treatment DCE-MRI from 1,506 patients with breast cancer who underwent NAC. It outperformed state-of-the-art models with an AUC of 0.73 (69.09% balanced accuracy, 75.27% sensitivity, 62.91% specificity) in the highly challenging task of predicting NAC response. Our framework could support early patient stratification towards more efficient treatments to improve personalized prognosis.

Keywords: Deep Learning, Spatiotemporal Transformer, Breast Tumor Segmentation, Pathologic Complete Response.

1 Introduction

Breast cancer is the most commonly diagnosed malignancy among women worldwide and accounts for more than 600,000 deaths annually [1]. Neo-adjuvant chemotherapy (NAC), chemotherapy administered before surgery, has been increasingly used to treat women with aggressive breast tumor subtypes, including those diagnosed with triple-negative and HER2-positive disease. While pathologic complete response (pCR) (i.e., the absence of invasive cancer in both the breast and axillary lymph nodes) to NAC has been strongly associated with improved disease-free survival, many women show partial or no positive response. In these cases, the delays introduced by ineffective treatment may contribute to decreased success rates in addition to the common adverse effects of chemotherapy. While clinical characteristics, molecular biomarkers, and other pathological features have been used to predict treatment response, they often fail to capture the complex interplay between cancer cells and the surrounding microenvironment, whose importance to determine NAC response is being increasingly recognized and has motivated the use of DCE-MRI to help diagnose and monitor cancer [2, 3] .

Several methods have achieved impressive results on pCR classification using both pre- and post-treatment DCE-MRI [4–6]. However, their reliance on post-treatment imaging limits their utility to predict treatment response. Unfortunately, while automated methods for pre-treatment pCR prediction may achieve high performance, they are only accurate in specific tumor subtypes, rely on expert-annotated tumoral and peritumoral volumes [7] or require pharmacokinetic modeling involving complex pre-processing and precise arterial input function selection [8], which not only affect reproducibility but make them impractical for large-scale clinical implementation.

Similar to other domains, the nnU-Net framework [9] has become a benchmark for tumor segmentation from DCE-MRI and has been recently used for pCR classification tasks prior to treatment [10–12]. However, the accuracy of pCR predictions based on these segmentations are typically reported under 60%, which suggests that the geometric information provided by tumor segmentation is insufficient to predict NAC response. One of the main challenges is that nnU-Net-based architectures require simplifying, summarizing or disregarding the temporal information of contrast uptake and washout in DCE-MRI sequences, which limit their ability to adequately characterize metabolic information and the local tumor environment [13].

Addressing current state-of-the-art challenges, we present a novel deep learning framework that can evaluate all the spatial and temporal information embedded in DCE-MRI sequences to predict pCR after NAC treatment in women with breast cancer. Unlike state-of-the-art methods, our network incorporates a new transformer-based architecture that learns 4D features from DCE-MRI by aggregating spatiotemporal embeddings, using exact phase acquisition times rather than sampling the temporal domain. This inclusion of all spatiotemporal information about contrast kinetic patterns, underlying tumor physiology, and tumor geometry enables the model to differentiate

aggressive washout behaviors from benign enhancement and ultimately yields higher predictive accuracy for treatment response than existing methods.

2 Materials and Methods

2.1 Dataset

A retrospective public dataset with pre-treatment DCE-MRI studies from 1,506 women (age 48.5 ± 10.4 years, range 21-77 years) with breast cancer prior to NAC from four sources was used: Duke-Breast-Cancer-MRI (DUKE, *N*=291), two datasets from Investigation of Serial studies to Predict Your Therapeutic Response with Imaging And moLecular analysis (ISPY 1, *N*=171 and 2, *N*=980), and Breast-MRI-NACT-Pilot (NACT, *N*=64) [14]. Each study contained one T1-weighted pre-contrast breast MRI and between two and five post-contrast images, totaling 7,926 images (in-plane resolution 0.72 ± 0.14 mm, slice thickness 1.87 ± 0.59 mm, sequence duration 524 ± 168 seconds). Segmentation masks of the primary breast tumor annotated by expert radiologists were available in addition to a bounding box for the breast location and NAC treatment response (whether the patient showed pCR or not). In addition, a spreadsheet of clinical variables was available including patient age, menopausal status, breast density, body mass index group, multifocal cancer status, anti-her2 therapy, estrogen present, progesterone present, her2 present, tumor subtype, and Nottingham grade [14]. 440 (29.5%) patients showed pCR after NAC and 1,066 (70.5%) did not.

Each image was cropped around the breast using available bounding-box data and resampled to isotropic spacing of $2 \times 2 \times 2$ mm^3, thus producing image sizes between $38 \times 51 \times 42$ and $103 \times 167 \times 105$ voxels. Each image sequence was represented as a 4D image with $T \times W \times D \times W$ voxels, where T is the number of contrast phase images, W their width, H their height, and D their depth.

2.2 Model Architecture

We propose the self-attention transformer-based architecture presented in Fig. 1 to estimate the probability of pCR to NAC from pre-treatment DCE-MRI phase sequences and tumor segmentations. The input to our model is a DCE-MRI image sequence with shape $T \times W \times H \times D$, and a binary segmentation image with shape $1 \times W \times H \times D$. Each DCE-MRI sequence i was spatially divided into N_i patches with fixed size of $64 \times 64 \times 64$ voxels, resulting in a $N_i \times T_i \times 64 \times 64 \times 64$ tensor, where T_i represents the number of longitudinal images in the temporal sequence (which is different between subjects). This reduces computational load by avoiding loading the full 4D image into memory at once. Then, a patch embedding encoder with five convolutional blocks was used to learn an embedding with 1,152 spatiotemporal features from each 4D patch with size $T_i \times 64 \times 64 \times 64$. The resulting embedding tensor was represented as a matrix with size $N_i T_i \times 1{,}152$.

We propose concatenating each patch embedding obtained from the DCE-MRI sequence with a new spatiotemporal encoding vector that represents its position in the spatiotemporal domain, unlike existing methods that reduce the dimensionality of data

or encode position and perform self-attention one dimension at a time. Our method simultaneously encodes all dimensions of the data and unravels high-dimensional tensors into a single sequence suitable for the attention mechanism. In our implementation, we used sinusoidal encodings [15], which decode single values into position encoding vectors of any desired size. We used four input values corresponding to the phase acquisition time t_i and the spatial patch indices $d_{i,j}, w_{i,j}, h_{i,j}$ within the image, where j represents a patch index and i the phase index in the DCE-MRI sequence. We empirically set the encoding vector size to be 288. We concatenated the four encoding vectors together, generating a unique, deterministic encoding vector for each patch with size 1,152. We then added each position encoding vector to its corresponding patch embedding. This spatiotemporal encoding approach allows our transformer-based architecture to more accurately identify temporal variations compared to traditional approaches that treat DCE-MRI phases as equally spaced channels.

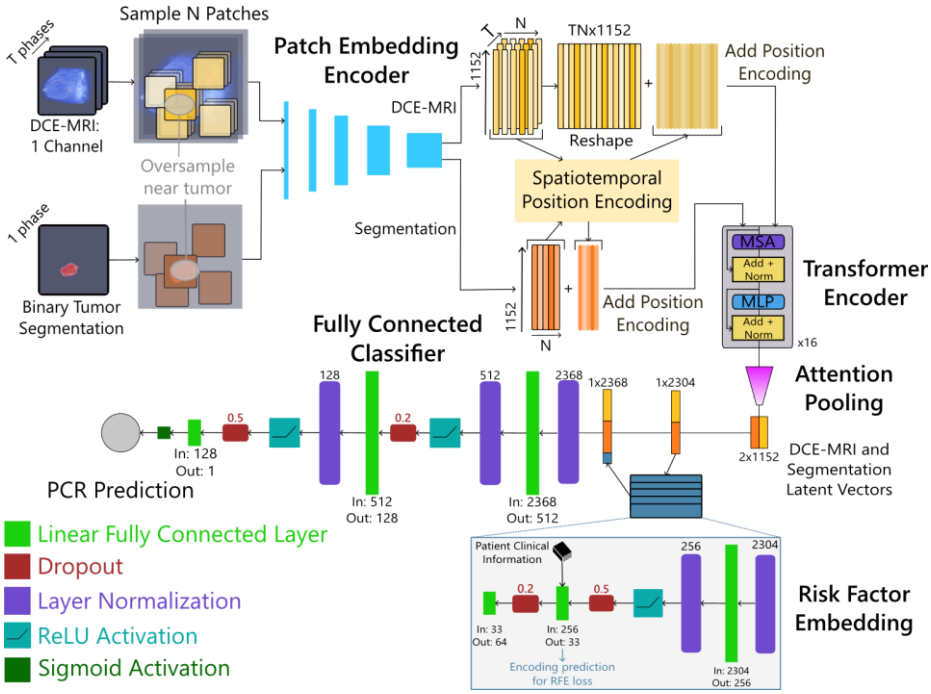


Fig. 1. Proposed spatiotemporal transformer-based architecture.

The sequence of patch embeddings with spatiotemporal encodings are fed to a transformer-encoder [15] with sixteen consecutive layers of 32 self-attention heads followed by a residual addition and layer normalization, a fully-connected feed-forward loop (two linear transformations with a ReLU activation in between) with another residual addition, and layer normalization (see [15] for details). This transformer-encoder accumulates both spatial and temporal information from all patches before an attention pooling layer [16], which creates a single spatiotemporal latent vector. To incorporate

tumor-specific information, the binary segmentation patches follow the same architecture, resulting in a second latent vector that is concatenated with the first as shown in Fig. 1. Since segmentation images do not include temporal information, they are encoding using the phase acquisition time of 0.

As shown in Fig. 1, an addition to the DCE-MRI image sequence, our framework incorporates a risk factor embedding with the following pre-treatment clinical variables that can help predict NAC response: age, menopausal status, breast density, body mass index group (underweight, normal, overweight, obese), multifocal cancer status, anti-her2 therapy status, estrogen present, progesterone present, her2 present, tumor subtype, and Nottingham grade. These pre-treatment clinical variables were transformed into a single feature vector of size 33 via one-hot encoding to produce a risk factor embedding for the model (see Fig. 1). Age was categorized into four groups (0-40, 41-50, 51-60, >61 years) as suggested by clinical experts. Since some or all of these features may not be available for a specific patient, our architecture incorporates a dedicated linear layer that predicts the one-hot encoded representation of these clinical factors from the spatiotemporal latent feature vector as shown in Fig. 1. A binary mask for each feature is used during inference to determine if the predicted or true clinical factors will be used for prediction. The clinical variables are predicted via an MLP-based architecture comprising layer normalization, a 256-feature linear module, a second layer normalization, ReLU activation, 50% dropout, and a final linear layer that outputs the 33-dimensional predicted clinical profile. When available, ground-truth encodings are substituted for the predicted variables via a binary mask. This 33-dimensional encoding is then mapped to a 64-feature embedding via 20% dropout and a linear module. This embedding is finally concatenated with the spatiotemporal latent feature vector and fed to a final linear classification module that consists of layer normalization, a 512-feature linear module, a second layer normalization, ReLU activation, 20% dropout, a 128-feature linear module, a third layer normalization, ReLU activation, 50% dropout, and a final single-feature linear module with a sigmoid activation to estimate the probability of pCR.

2.3 Training

We propose training our neural network using a loss function with two terms. The first term quantifies its pCR prediction accuracy using a binary cross-entropy formulation that balances the contribution between positive and negative samples in our datasets as

$$\mathcal{L}_{CLS}(\hat{\boldsymbol{y}}, \boldsymbol{y}|w) = \sum_{i=1}^{N} w\boldsymbol{y_i}\mathrm{Log}(\hat{\boldsymbol{y}}_i) + (1-\boldsymbol{y_i})(1-\mathrm{Log}(\hat{\boldsymbol{y}}_i)), \quad (1)$$

where $\hat{\boldsymbol{y}}$ is the predicted probability, $\boldsymbol{y}$ the true label, N the number of patients, and w is the weighting parameter balancing the contribution between the two classes.

The second term quantifies the performance of the risk factor embedding module at predicting clinical factors using a binary cross-entropy formulation as

$$\mathcal{L}_{RFE}(\hat{\boldsymbol{R}}, \boldsymbol{R}) = \sum_{i=1}^{N} \frac{1}{||\mathrm{V}_i||} \sum_{j \in \mathrm{V}_i} \boldsymbol{R}_{i,j}\mathrm{Log}\left(\hat{\boldsymbol{R}}_{i,j}\right) + \left(1-\boldsymbol{R}_{i,j}\right)\left(1-\mathrm{Log}\left(\hat{\boldsymbol{R}}_{i,j}\right)\right), \quad (2)$$

where $\widehat{\boldsymbol{R}}$ is the predicted one-hot risk factor encoding, $\boldsymbol{R}$ the true encoding, and V_i is the set of available clinical variables for a patient. During training, this module is optimized against the ground-truth clinical data, with backpropagation occurring only for known features by masking out missing patient information from the loss calculation. Hence, the loss function minimized during training is:

$$\mathcal{L}(\hat{\boldsymbol{y}}, \boldsymbol{y}, \widehat{\boldsymbol{R}}, \boldsymbol{R}|w) = \lambda_{CLS}\mathcal{L}_{CLS}(\hat{\boldsymbol{y}}, \boldsymbol{y}|w) + \lambda_{RFE}\mathcal{L}_{RFE}(\widehat{\boldsymbol{R}}, \boldsymbol{R}) \quad (3)$$

where λ_{CLS} and λ_{RFE} are hyperparameters balancing the contribution of both loss terms.

Training was performed using the Adam optimizer with a decoupled weight decay of 0.01, and with $\beta_1 = 0.90$ and $\beta_2 = 0.999$ for the exponential decay of the first and second moments, respectively. The model ran for 800 epochs with a batch size of 20 before stopping after no improvement for 100 epochs in the validation dataset. A learning rate scheduler with half-cosine decay from 5E-5 to 1E-6 was used.

3 Experiments and Results

We divided our data into training (70%), validation (10%), and testing (20%) splits, stratifying by data source, tumor subtype, and pCR status. We empirically set the hyperparameters as $\lambda_{CLS} = 3$ and $\lambda_{RFE} = 1$. We set the class balance weighting parameter $w = 2.4$, equal to the ratio of the negative pCR samples to positive pCR samples present in the data set. During training, to prioritize image regions with tumor information, each image was sampled three different times at every epoch. This helped the model focus on regions near the tumor, which contain important anatomical information about the tumor microenvironment. In two of the samplings, patches were retrieved around the tumor with a randomized overlap probability drawn from a normal distribution with 50% mean and 15% standard deviation. In the third sampling, patches were sampled outside the tumor region to incorporate context from non-cancerous tissue. During validation and testing, the entire image was sampled using a randomized path order. The model was trained on an NVIDIA A6000 GPU.

The classification results are compared with other state-of-the-art methods on the same dataset in Table 1. Additionally, Table 2 shows stratified results by age, menopausal status and data source. The reported balanced accuracy was obtained with the probability threshold that maximized the Youden's index.

Table 1. Model Results Comparison

Model	AUC	Bal. Acc.
Proposed	0.73	0.69
Radiomic Self-Normalizing Network [12]	0.59	0.57
XGBoost [10]	0.53	0.51
Gradient Boosted Decision Tree [11]	0.52	0.51

Table 2. Stratified Proposed Model Results (Test Split)

Demographic	# Patients in Group	Bal. Acc.	Sens.	Spec.
Overall	306	0.69	0.75	0.63
Age Group (Years)				
0-40	77	0.70	0.77	0.63
41-50	120	0.73	0.82	0.63
51-60	78	0.66	0.67	0.65
61+	30	0.56	0.56	0.57
Menopausal Status				
Post	62	0.66	0.60	0.71
Pre	130	0.71	0.82	0.59
Unknown	114	0.69	0.76	0.62
Data Source				
DUKE	91	0.72	0.73	0.71
ISPY1	67	0.72	0.78	0.66
ISPY2	131	0.66	0.76	0.56
NACT	17	0.45	0.33	0.57

4 Discussion

We presented a new 4D transformer-based framework that can effectively incorporate both spatial and temporal information from variable length DCE-MRI sequences to predict response to NAC treatment in women with breast cancer. Unlike other state-of-the-art models [10–12], our novel 4D transformer-based architecture conceptualizes the DCE-MRI acquisition as a continuous temporal process rather than a sequence of independent channels, and does not require temporal sampling or alignment, to accurately characterize the dynamics of the tumor microenvironment and its surrounding tissue.

Our results show an improved performance over other state-of-the-art methods [10–12]. Specifically, our improved balanced accuracy and AUC suggest that both the dynamic information embedded in the DCE-MRI sequence and the surrounding tissues provide valuable information for predicting therapeutic response. With the proposed design, in addition to contrast uptake and washout patterns, our model learns to capture long-range spatial dependencies, which prior work shows is representative of the tumor microenvironment and relevant to treatment response [17, 18]. Hence, unlike existing approaches, our 4D transformer-based framework can simultaneously evaluate the tumor core, peritumoral rim, and surrounding structures for an improved evaluation.

It is important to note that, although a few works report higher performance metrics than our proposed framework, they typically rely on additional imaging data and/or

expert manual annotations of volumes of interest for every patient [7], which is highly impractical for large-scale clinical deployment. Moreover, although methods using post-treatment DCE-MRI sequences can provide higher accuracy identifying pCR after NAC treatment, their reliance on post-treatment data limits their utility to identify the best course of treatment for a patient. Separately, other methods rely on the use of Tofts [19] model-derived kinetic parameters, which require measurements of arterial input function in addition to precise and consistent T1 mapping, which are notoriously challenging to standardize across centers. Unlike all those approaches, our model can predict treatment response using multi-institutional information that is generally available before treatment and incorporates a risk factor embedding that provides versatility if any clinical variable may not be available for a given patient. Moreover, our architecture utilizes multi-head self-attention to learn a more flexible representation of hemodynamics without making strong assumptions [8], unlike traditional pharmacokinetic models that rely on a fixed mathematical structures to interpret contrast exchange.

Our study has several limitations. First, transformers typically require large training datasets for stability. Indeed, our results show higher performance in larger patient groups. While expanding public dataset diversity [14] remains critical for equitable clinical translation, the increased performance of our spatiotemporal model compared to other state-of-the-art approaches shows the benefits of our novel design on existing public datasets. Our model also used $2 \times 2 \times 2$ mm^3 voxel size to increase memory efficiency at the expense of fine details but still achieved state-of-the-art performance. Another potential limitation is our multi-step processing approach. Since recent tumor segmentation methods have shown great performance in MRI [10–12], we decided to leverage existing segmentations and focus our architecture in the more complex task of capturing complex local dynamic information from variable-length DCE-MRI sequences. Although our model does not fully eliminate the need for tumor segmentations, its extension to generate tumor segmentations using a multi-task framework is part of our future work. Finally, no experiments have explicitly tested the isolated contribution of our risk-factor submodule and the proposed 4D spatiotemporal convolutions as compared to standard 3D convolutions. While the first remains part of future work, the use of standard 3D convolutions would prevent efficiently leveraging temporal relationships to reduce data dimensionality, which would increase memory needs during training and require computational resources that are not available to our team. Despite these limitations, our spatiotemporal formulation showed consistent performance gains over the compared frameworks.

The architecture proposed in this study can effectively consider both spatial and temporal information in variable-length longitudinal 3D image sequences. This allows capturing complex dynamic patterns to characterize differential post-contrast local tissue characteristics that may be relevant for diagnosis and prognosis. We showed increased performance compared to state-of-the-art methods in the prediction of response to NAC treatment in women with breast cancer. If validated prospectively across diverse clinical settings, this framework could enable earlier identification of patients unlikely to achieve a pathologic complete response, supporting more personalized treatment selection before initiation of neoadjuvant chemotherapy for women with breast cancer.

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